

- » Introducing the internet
- » Familiarizing yourself with TCP/IP standards
- » Figuring out how TCP/IP lines up with the OSI Reference Model
- » Discovering important TCP/IP applications

Chapter 2

TCP/IP and the Internet

Many years ago, Transmission Control Protocol/Internet Protocol (TCP/IP) was known primarily as the protocol of the internet. The biggest challenge of getting a local area network (LAN) connected to the internet was figuring out how to mesh TCP/IP with the proprietary protocols that were the basis of the LANs — most notably Internetwork Packet Exchange/Sequenced Packet Exchange (IPX/SPX) used by Novell networks and NetBIOS Extended User Interface (NetBEUI) used by Microsoft networks.

Eventually, both IPX/SPX and NetBIOS gave way to TCP/IP as the basis for local area networking, eliminating the challenge of translating IPX/SPX or NetBEUI to TCP/IP. As a result, TCP/IP is not just the protocol of the internet now, but it's also the protocol on which most LANs are based.

This chapter is a gentle introduction to the internet in general and the TCP/IP suite of protocols in particular. After I get the introductions out of the way, you'll be able to focus more in-depth on the detailed TCP/IP information given in the remaining chapters of Book 2.

What Is the Internet?

The Goliath of all computer networks, the internet links hundreds of millions of computer users throughout the world. Strictly speaking, the internet is a network of networks. It consists of hundreds of thousands of separate computer networks, all interlinked, so that a user on any of those networks can reach out and potentially touch a user on any of the other networks. This network of networks connects more than a billion computers to each other. (That's right, *billion* with a *b*.)

One of the official documents (RFC 2026) of the Internet Engineering Task Force (IETF) defines the internet as “a loosely organized international collaboration of autonomous, interconnected networks.” Broken down piece by piece, this definition encompasses several key aspects of what the internet is:

» **Loosely organized:** No single organization has authority over the internet. As a result, the internet is not highly organized. Online services, such as America Online or MSN, are owned and operated by individual companies that control exactly what content appears on the service and what software can be used with the service. No one exercises that kind of control over the internet. As a result, you can find just about any kind of material imaginable on the internet. No one guarantees the accuracy of information that you find on the internet, so you have to be careful as you work your way through the labyrinth.



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JUST HOW BIG IS THE INTERNET?

Because the internet is not owned or controlled by any one organization, no one knows how big the internet really is. Several organizations do attempt to periodically determine the size of the internet, including the Internet Systems Consortium (ISC), which completed its last survey in January 2019 and found that well over a billion host computers are connected to the internet. The first year the ISC did the survey (1993), it found only 1.3 million host computers. It passed 10 million hosts in 1996, 100 million hosts in 2000, and edged over 1 billion hosts in 2014.

Unfortunately, no one knows how many actual users are on the internet. Each host can support a single user — or in the case of domains, dozens, hundreds, thousands, or perhaps even millions of users. No one really knows.

In fact, the ISC gave up on trying to count the number of hosts on the internet. The January 2019 survey was its last. If you're interested, you can check its historical survey data at www.isc.org/network/survey.

- » **International:** More than 300 countries are represented on the internet, from Ascension Island to Zimbabwe.
- » **Collaboration:** The internet exists only because many different organizations cooperate to provide the services and support needed to sustain it. For example, much of the software that drives the internet is open source software that's developed collaboratively by programmers throughout the world, who constantly work to improve the code.
- » **Autonomous:** The internet community respects that organizations that join the internet are free to make their own decisions about how they configure and operate their networks. Although legal issues sometimes boil up, for the most part, each player on the internet operates independently.
- » **Interconnected:** The whole key to the internet is the concept of *interconnection*, which uses standard protocols that enable networks to communicate with each other. Without the interconnection provided by the TCP/IP protocol, the internet would not exist.
- » **Networks:** The internet would be completely unmanageable if it consisted of half a billion individual users, all interconnected. That's why the internet is often described as a network of networks. Most individual users on the internet don't access the internet directly. Instead, they access the internet indirectly through another network, which may be a LAN in a business or academic environment, or a dialup or broadband network provided by an internet service provider (ISP). In each case, however, the users of the local network access the internet via a gateway IP router.

The internet is composed of several distinct types of networks: government agencies, such as the Library of Congress and the White House; military sites (did you ever see *War Games* or any of the *Terminator* movies?); educational institutions, such as universities and colleges (and their libraries); businesses, such as Microsoft and IBM; ISPs, which allow individuals to access the internet; and commercial online services, such as America Online and MSN.

A Little Internet History

The internet has a fascinating history, if such things interest you. There's no particular reason why you should be interested in such things, of course, except that a superficial understanding of how the internet got started may help you to understand and cope with the way this massive computer network exists today. So here goes.

The internet traces its beginnings back to a small network called ARPANET, built by the Department of Defense in 1969 to link defense installations. ARPANET soon expanded to include not only defense installations but universities as well. In the 1970s, ARPANET was split into two networks: one for military use (renamed MIL-NET) and the original ARPANET (for nonmilitary use). The two networks were connected by a networking link called the *Internet Protocol* (IP), so called because it allowed communication between two networks.

The good folks who designed IP had the foresight to realize that soon, more than two networks would want to be connected. In fact, they left room for tens of thousands of networks to join the game, which is a good thing because it wasn't long before the internet began to take off.

By the mid-1980s, ARPANET was beginning to reach the limits of what it could do. Enter the National Science Foundation (NSF), which set up a nationwide network designed to provide access to huge *supercomputers*, those monolithic computers used to discover new prime numbers and calculate the orbits of distant galaxies. The supercomputers were never put to much use, but the network that was put together to support the supercomputers — NSFNET — was used. In fact, NSFNET replaced ARPANET as the new backbone for the internet.

Then, out of the blue, it seemed as if the whole world became interested in the internet. Stories about it appeared in *Time* and *Newsweek*. Any company that had “dot com” in its name practically doubled in value every month. Al Gore claimed he invented the internet (well, not really — but he was *accused* of taking credit). The Net began to grow so fast that even NSFNET couldn't keep up, so private commercial networks got into the game. The size of the internet nearly doubled every year for most of the 1990s. Then, in the first few years of the millennium, the growth rate slowed a bit. However, the internet still seems to be growing at the phenomenal rate of about 30 to 50 percent per year, and who knows how long this dizzying rate of growth will continue.

TCP/IP Standards and RFCs

The TCP/IP protocol standards that define how the internet works are managed by the IETF. However, the IETF doesn't impose standards. Instead, it simply oversees the process by which ideas are developed into agreed-upon standards.

An internet standard is published in the Request for Comments (RFC) document. When a document is accepted for publication, it is assigned an RFC number by the

IETF. The RFC is then published. After it's published, an RFC is never changed. If a standard is enhanced, the enhancement is covered in a separate RFC.

Thousands of RFCs are available from the IETF's RFC Editor website (www.rfc-editor.org). The oldest RFC is RFC 0001, published in April 1969. It describes how the host computers communicated with each other in the original ARPANET. As of this writing, the most recent proposed standard is RFC 9565, entitled "An Update to the tcpControlBits IP Flow Information Export (IPFIX) Information Element."

Not all RFCs represent internet standards. The following paragraphs summarize the various types of RFC documents:

- » **Internet Standards Track:** This type of RFC represents an internet standard. Standards Track RFCs have one of three maturity levels, as described in Table 2-1. An RFC enters circulation with Proposed Standard status but may be elevated to Draft Standard status — and, ultimately, to Internet Standard status.
- » **Experimental specifications:** These are a result of research or development efforts. They're not intended to be standards, but the information they contain may be of use to the internet community.
- » **Informational specifications:** These simply provide general information for the internet community.
- » **Historic specifications:** These RFCs have been superseded by a more recent RFC and are thus considered obsolete.
- » **Best current practice (BCP):** RFCs are documents that summarize the consensus of the internet community's opinion on the best way to perform an operation or procedure. BCPs are guidelines, not standards.
- » **April Fools:** It's a long-standing tradition to post amusing RFCs on April Fools' Day. One of my favorites is RFC 1149, "A Standard for the Transmission of IP Datagrams on Avian Carriers." The specification calls for IP datagrams to be written in hexadecimal on scrolls of paper and secured to "avian carriers" with duct tape.

The most recent April 1 RFC at the time of this writing is RFC 9564, "Faster Than Light Speed Protocol (FLIP)." It defines a protocol in which the receiver uses AI technology to predict the contents of packets before they have even been sent, thereby achieving faster-than-light internet speeds.

Table 2-2 summarizes the RFCs that apply to the key internet standards described in this book.

TABLE 2-1 **Maturity Levels for Internet Standards Track RFCs**

Maturity Level	Description
Proposed Standard	Generally stable, have resolved known design choices, are believed to be well understood, have received significant community review, and appear to enjoy enough community interest to be considered valuable.
Draft Standard	Well understood and known to be quite stable. At least two interoperable implementations must exist, developed independently from separate code bases. The specification is believed to be mature and useful.
Internet Standard	Have been fully accepted by the internet community as highly mature and useful standards.

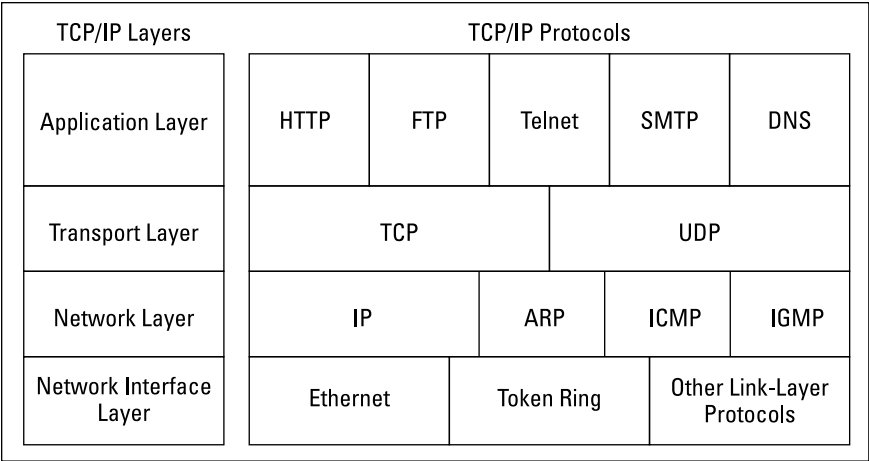
TABLE 2-2 **RFCs for Key Internet Standards**

RFC	Date	Description
768	August 1980	User Datagram Protocol (UDP)
791	September 1981	Internet Protocol (IP)
792	September 1981	Internet Control Message Protocol (ICMP)
793	September 1981	Transmission Control Protocol (TCP)
826	November 1982	Ethernet Address Resolution Protocol (ARP)
950	August 1985	Internet Standard Subnetting Procedure
959	October 1985	File Transfer Protocol (FTP)
1034	November 1987	Domain Names — Concepts and Facilities (DNS)
1035	November 1987	Domain Names — Implementation and Specification (DNS)
1939	May 1996	Post Office Protocol Version 3 (POP3)
2131	March 1997	Dynamic Host Configuration Protocol (DHCP)
3376	November 1997	Internet Group Management Protocol (IGMP) (Updates RFC 2236 and 1112)
5321	October 2008	Simple Mail Transfer Protocol (SMTP)
7230 through 7235	June 2014	Hypertext Transfer Protocol – HTTP/1.1

The TCP/IP Protocol Framework

Like the seven-layer OSI Reference Model, TCP/IP protocols are based on a layered framework. TCP/IP has four layers, as shown in Figure 2-1. These layers are described in the following sections.

FIGURE 2-1:
The four layers
of the TCP/IP
framework.



Network interface layer

The lowest level of the TCP/IP architecture is the network interface layer. It corresponds to the OSI physical and data link layers. You can use many different TCP/IP protocols at the network interface layer, including Ethernet and Token Ring for LANs and protocols such as X.25, Frame Relay, and ATM for wide area networks (WANs).

The network interface layer is assumed to be unreliable.

Network layer

The network layer is where data is addressed, packaged, and routed among networks. Several important internet protocols operate at the network layer:

- » **Internet Protocol (IP):** A routable protocol that uses IP addresses to deliver packets to network devices. IP is an intentionally unreliable protocol, so it doesn't guarantee delivery of information.
- » **Address Resolution Protocol (ARP):** Resolves IP addresses to hardware Media Access Control (MAC) addresses, which uniquely identify hardware devices.
- » **Internet Control Message Protocol (ICMP):** Sends and receives diagnostic messages. ICMP is the basis of the ubiquitous ping command.
- » **Internet Group Management Protocol (IGMP):** Used to multicast messages to multiple IP addresses at once.

Transport layer

The transport layer is where sessions are established and data packets are exchanged between hosts. Two core protocols are found at this layer:

- » **Transmission Control Protocol (TCP):** Provides reliable connection-oriented transmission between two hosts. TCP establishes a session between hosts, and then ensures delivery of packets between the hosts.
- » **User Datagram Protocol (UDP):** Provides connectionless, unreliable, one-to-one or one-to-many delivery.

Application layer

The application layer of the TCP/IP model corresponds to the session, presentation, and application layers of the OSI Reference Model. A few of the most popular application layer protocols are

- » **Hypertext Transfer Protocol (HTTP):** The core protocol of the World Wide Web.
- » **File Transfer Protocol (FTP):** A protocol that enables a client to send and receive complete files from a server.
- » **Telnet:** The protocol that lets you connect to another computer on the internet in a terminal emulation mode.
- » **Simple Mail Transfer Protocol (SMTP):** One of several key protocols that are used to provide email services.
- » **Domain Name System (DNS):** The protocol that allows you to refer to other host computers by using names rather than numbers.